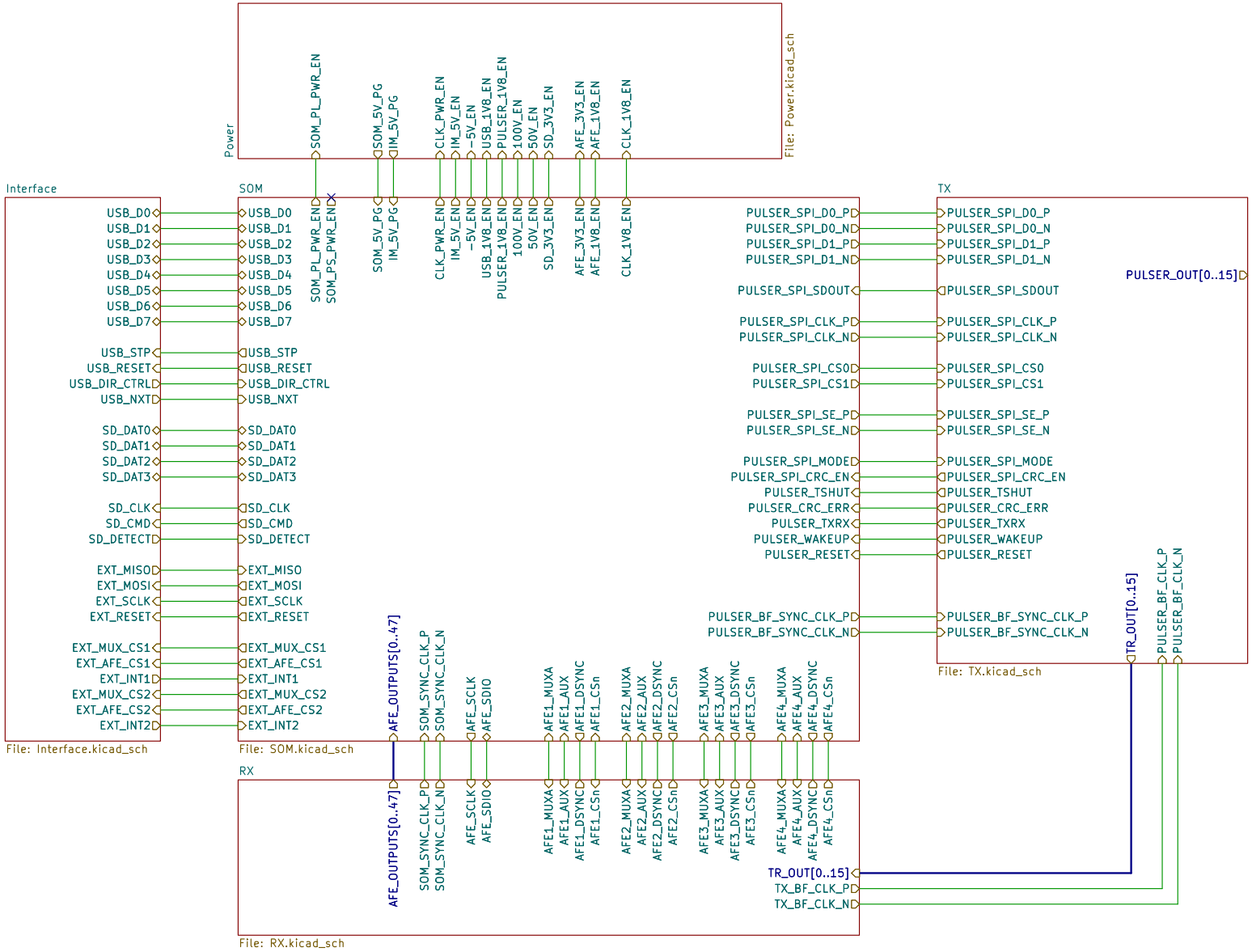
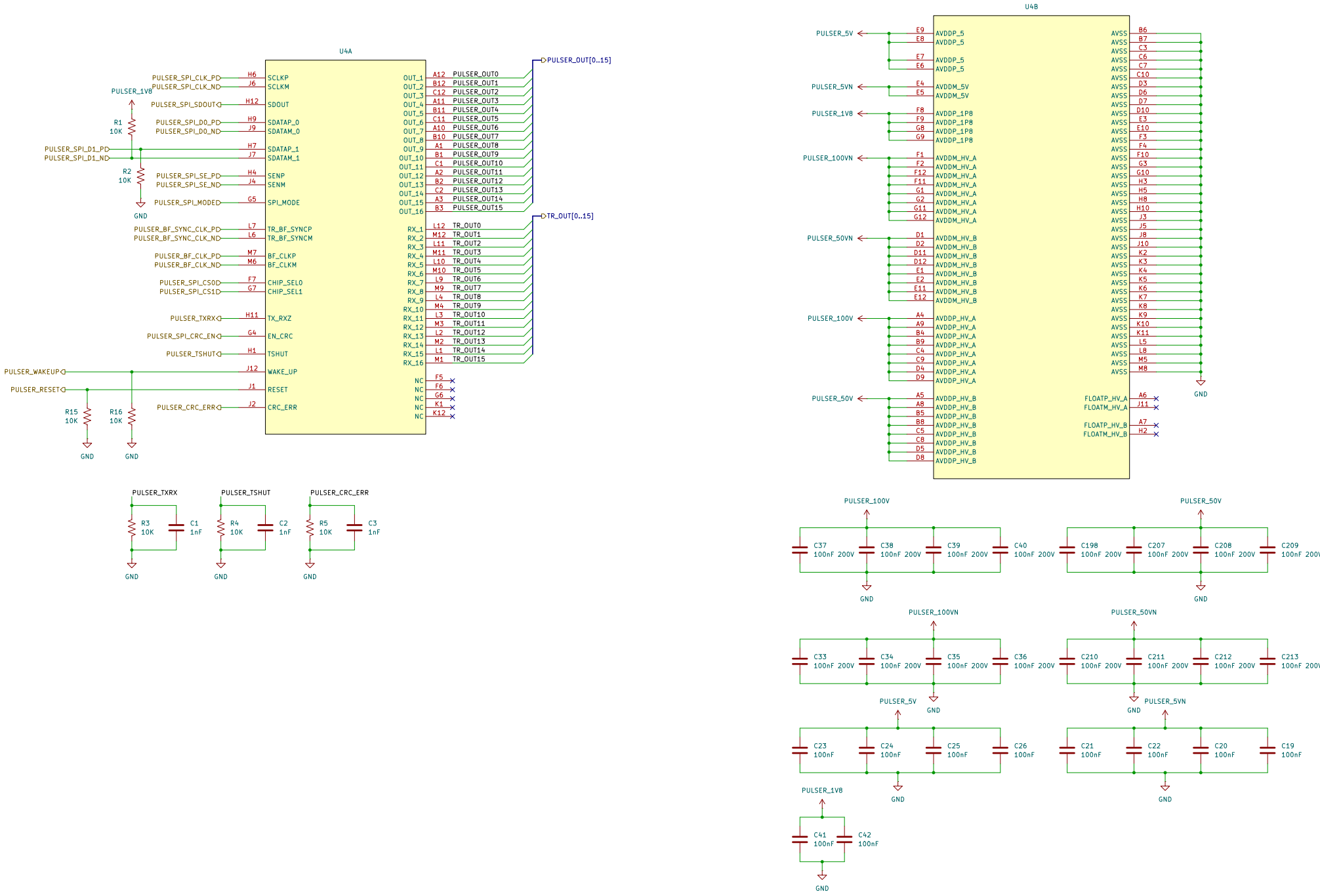
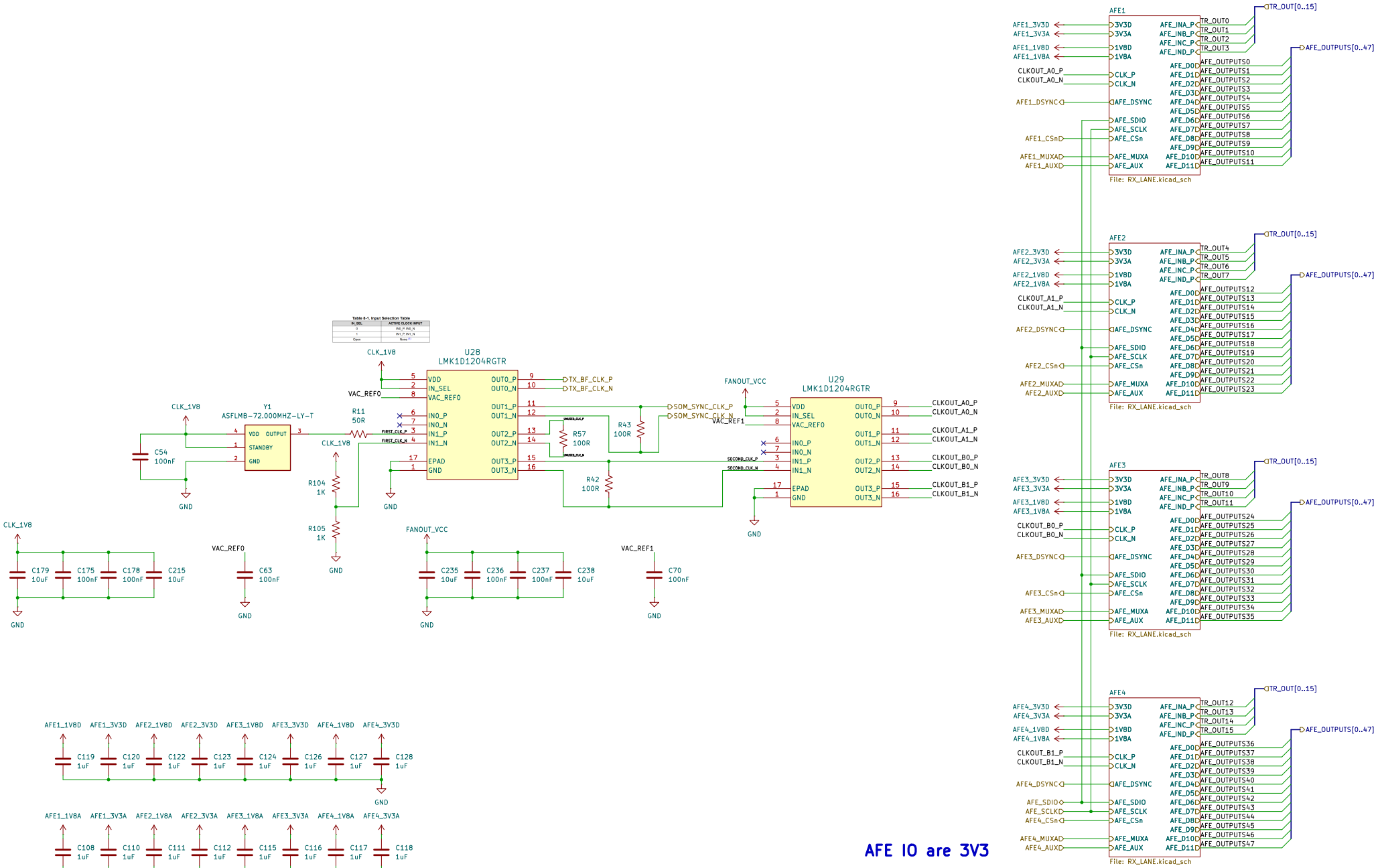
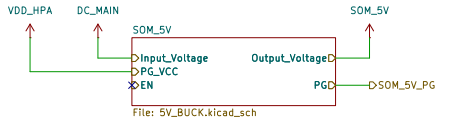

Power
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SOM_PL_PWR_EN
SOM_5V_PG
IM_5V_PG
CLK_PWR_EN
IM_5V_EN
-5V_EN
USB_1V8_EN
PULSER_1V8_EN
100V_EN
50V_EN
SD_3V3_EN
AFE_3V3_EN
AFE_1V8_EN
CLK_1V8_EN
Interface
USB_D0
USB_D1
USB_D2
USB_D3
USB_D4
USB_D5
USB_D6
USB_D7
USB_STP
USB_RESET
USB_DIR_CTRL
USB_NXT
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_CLK
SD_CMD
SD_DETECT
EXT_MISO
EXT_MOSI
EXT_SCLK
EXT_RESET
EXT_MUX_CS1
EXT_AFE_CS1
EXT_INT1
EXT_MUX_CS2
EXT_AFE_CS2
EXT_INT2
File: Interface.kicad_sch
SOM
SOM_PS_PWR_EN
AFE_OUTPUTS[0..47]
SOM_SYNC_CLK_P
SOM_SYNC_CLK_N
AFE_SCLK
AFE_SDIO
AFE1_MUXA
AFE1_AUX
AFE1_DSYNC
AFE1_CSn
AFE2_MUXA
AFE2_AUX
AFE2_DSYNC
AFE2_CSn
AFE3_MUXA
AFE3_AUX
AFE3_DSYNC
AFE3_CSn
AFE4_MUXA
AFE4_AUX
AFE4_DSYNC
AFE4_CSn
File: SOM.kicad_sch
TX
PULSER_SPI_D0_P
PULSER_SPI_D0_N
PULSER_SPI_D1_P
PULSER_SPI_D1_N
PULSER_SPI_SDOUT
PULSER_SPI_CLK_P
PULSER_SPI_CLK_N
PULSER_SPI_CS0
PULSER_SPI_CS1
PULSER_SPI_SE_P
PULSER_SPI_SE_N
PULSER_SPI_MODE
PULSER_SPI_CRC_EN
PULSER_TSHUT
PULSER_CRC_ERR
PULSER_TXRX
PULSER_WAKEUP
PULSER_RESET
PULSER_BF_SYNC_CLK_P
PULSER_BF_SYNC_CLK_N
PULSER_OUT[0..15]
TR_OUT[0..15]
PULSER_BF_CLK_P
PULSER_BF_CLK_N
File: TX.kicad_sch
RX
TR_OUT[0..15]
TX_BF_CLK_P
TX_BF_CLK_N
File: RX.kicad_sch


U4A
PULSER_SPI_CLK_PD
PULSER_SPI_CLK_ND
PULSER_1V8
PULSER_SPI_SDOUT
R1
10K
PULSER_SPI_D0_PD
PULSER_SPI_D0_ND
PULSER_SPI_D1_PD
PULSER_SPI_D1_ND
R2
10K
PULSER_SPI_SE_PD
PULSER_SPI_SE_ND
PULSER_SPI_MODE
GND
PULSER_BF_SYNC_CLK_PD
PULSER_BF_SYNC_CLK_ND
PULSER_BF_CLK_PD
PULSER_BF_CLK_ND
PULSER_SPI_CS0
PULSER_SPI_CS1
PULSER_TXRX
PULSER_SPI_CRC_EN
PULSER_TSHUT
PULSER_WAKEUP
PULSER_RESET
PULSER_CRC_ERR
R15
10K
R16
10K
GND
GND
SCLKP
SCLKM
SDOUT
SDATAP_0
SDATAM_0
SDATAP_1
SDATAM_1
SENP
SENM
SPI_MODE
TR_BF_SYNCP
TR_BF_SYNCM
BF_CLKP
BF_CLKM
CHIP_SEL0
CHIP_SEL1
TX_RXZ
EN_CRC
TSHUT
WAKE_UP
RESET
CRC_ERR
PULSER_OUT[0..15]
PULSER_OUT0
PULSER_OUT1
PULSER_OUT2
PULSER_OUT3
PULSER_OUT4
PULSER_OUT5
PULSER_OUT6
PULSER_OUT7
PULSER_OUT8
PULSER_OUT9
PULSER_OUT10
PULSER_OUT11
PULSER_OUT12
PULSER_OUT13
PULSER_OUT14
PULSER_OUT15
TR_OUT[0..15]
TR_OUT0
TR_OUT1
TR_OUT2
TR_OUT3
TR_OUT4
TR_OUT5
TR_OUT6
TR_OUT7
TR_OUT8
TR_OUT9
TR_OUT10
TR_OUT11
TR_OUT12
TR_OUT13
TR_OUT14
TR_OUT15
NC
NC
NC
NC
NC
PULSER_TXRX
R3
10K
C1
1nF
GND
PULSER_TSHUT
R4
10K
C2
1nF
GND
PULSER_CRC_ERR
R5
10K
C3
1nF
GND
U4B
PULSER_5V
PULSER_5VN
PULSER_1V8
PULSER_100VN
PULSER_50VN
PULSER_100V
PULSER_50V
AVDDP_5
AVDDM_5V
AVDDP_1P8
AVDDM_HV_A
AVDDM_HV_B
AVDDP_HV_A
AVDDP_HV_B
AVSS
FLOATP_HV_A
FLOATM_HV_A
FLOATP_HV_B
FLOATM_HV_B
GND
PULSER_100V
C37
100nF 200V
C38
100nF 200V
C39
100nF 200V
C40
100nF 200V
GND
PULSER_50V
C198
100nF 200V
C207
100nF 200V
C208
100nF 200V
C209
100nF 200V
GND
PULSER_100VN
C33
100nF 200V
C34
100nF 200V
C35
100nF 200V
C36
100nF 200V
GND
PULSER_50VN
C210
100nF 200V
C211
100nF 200V
C212
100nF 200V
C213
100nF 200V
GND
PULSER_5V
C23
100nF
C24
100nF
C25
100nF
C26
100nF
GND
PULSER_5VN
C21
100nF
C22
100nF
C20
100nF
C19
100nF
GND
PULSER_1V8
C41
100nF
C42
100nF
GND

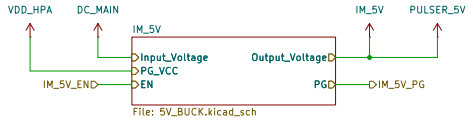

Table 8-1. Input Selection Table
IN_SEL
ACTIVE CLOCK INPUT
0
IN0_P, IN0_N
1
IN1_P, IN1_N
Open
None(1)
U28
LMK1D1204RGTR
U29
LMK1D1204RGTR
VDD
IN_SEL
VAC_REF0
IN0_P
IN0_N
IN1_P
IN1_N
EPAD
GND
OUT0_P
OUT0_N
OUT1_P
OUT1_N
OUT2_P
OUT2_N
OUT3_P
OUT3_N
CLK_1V8
VAC_REF0
VAC_REF1
FANOUT_VCC
Y1
ASFLMB-72.000MHZ-LY-T
VDD
OUTPUT
STANDBY
GND
C54
100nF
R11
50R
R104
1K
R105
1K
R57
100R
R43
100R
R42
100R
FIRST_CLK_P
FIRST_CLK_N
SECOND_CLK_P
SECOND_CLK_N
TX_BF_CLK_P
TX_BF_CLK_N
SOM_SYNC_CLK_P
SOM_SYNC_CLK_N
CLKOUT_A0_P
CLKOUT_A0_N
CLKOUT_A1_P
CLKOUT_A1_N
CLKOUT_B0_P
CLKOUT_B0_N
CLKOUT_B1_P
CLKOUT_B1_N
C179
10uF
C175
100nF
C178
100nF
C215
10uF
C63
100nF
C235
10uF
C236
100nF
C237
100nF
C238
10uF
C70
100nF
AFE1_1V8D
AFE1_3V3D
AFE2_1V8D
AFE2_3V3D
AFE3_1V8D
AFE3_3V3D
AFE4_1V8D
AFE4_3V3D
C119
1uF
C120
1uF
C122
1uF
C123
1uF
C124
1uF
C126
1uF
C127
1uF
C128
1uF
AFE1_1V8A
AFE1_3V3A
AFE2_1V8A
AFE2_3V3A
AFE3_1V8A
AFE3_3V3A
AFE4_1V8A
AFE4_3V3A
C108
1uF
C110
1uF
C111
1uF
C112
1uF
C115
1uF
C116
1uF
C117
1uF
C118
1uF
AFE1
AFE2
AFE3
AFE4
3V3D
3V3A
1V8D
1V8A
CLK_P
CLK_N
AFE_DSYNC
AFE_SDIO
AFE_SCLK
AFE_CSn
AFE_MUXA
AFE_AUX
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
TR_OUT0
TR_OUT1
TR_OUT2
TR_OUT3
TR_OUT4
TR_OUT5
TR_OUT6
TR_OUT7
TR_OUT8
TR_OUT9
TR_OUT10
TR_OUT11
TR_OUT12
TR_OUT13
TR_OUT14
TR_OUT15
TR_OUT[0..15]
AFE_OUTPUTS[0..47]
File: RX_LANE.kicad_sch
AFE IO are 3V3
GND

Main SOM 5V Supply



Power Rail Name	Supported Voltage Range	Maximum Current	Description
V_{CC_SOM}	5V (4.75V – 5.25V) 50 mV p-p maximum noise	4A	Main power input to the SOM. Supplies power to on-board power regulators. Total of 4A across connectors. SOM will consume a maximum of 3A.
V_{CC_BATT}	1.20 – 1.90V	100 nA – 500 nA	Coin cell battery input for the RTC
V_{CCO_HPA}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 66
V_{CCO_HPB}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 65
V_{CCO_HPC}	1.00V – 1.80V	1.0A	Voltage rail for HPIO bank 64
V_{CCO_HDA}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 45
V_{CCO_HDB}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 43
V_{CCO_HDC}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 44

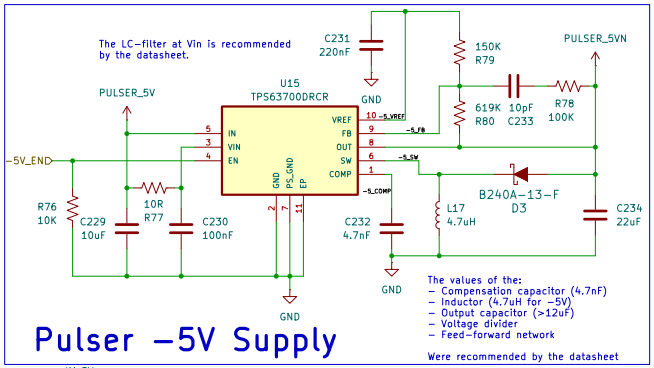
Pulser + Intermediate 5V Supply

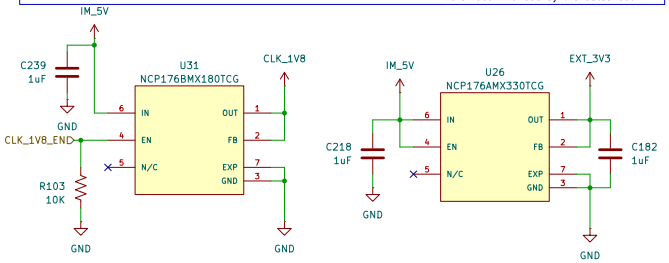


$I_{MODIO18}$	Full channel mode		130	mA
$I_{MODIO33}$	Full channel mode		130	mA
$I_{DVDDIO18}$			22	mA
$I_{DVDDIO33}$			2	mA

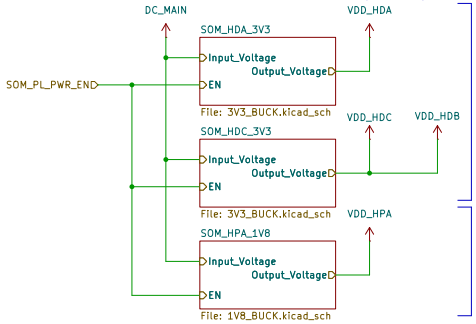
^ x4 = 1.2A

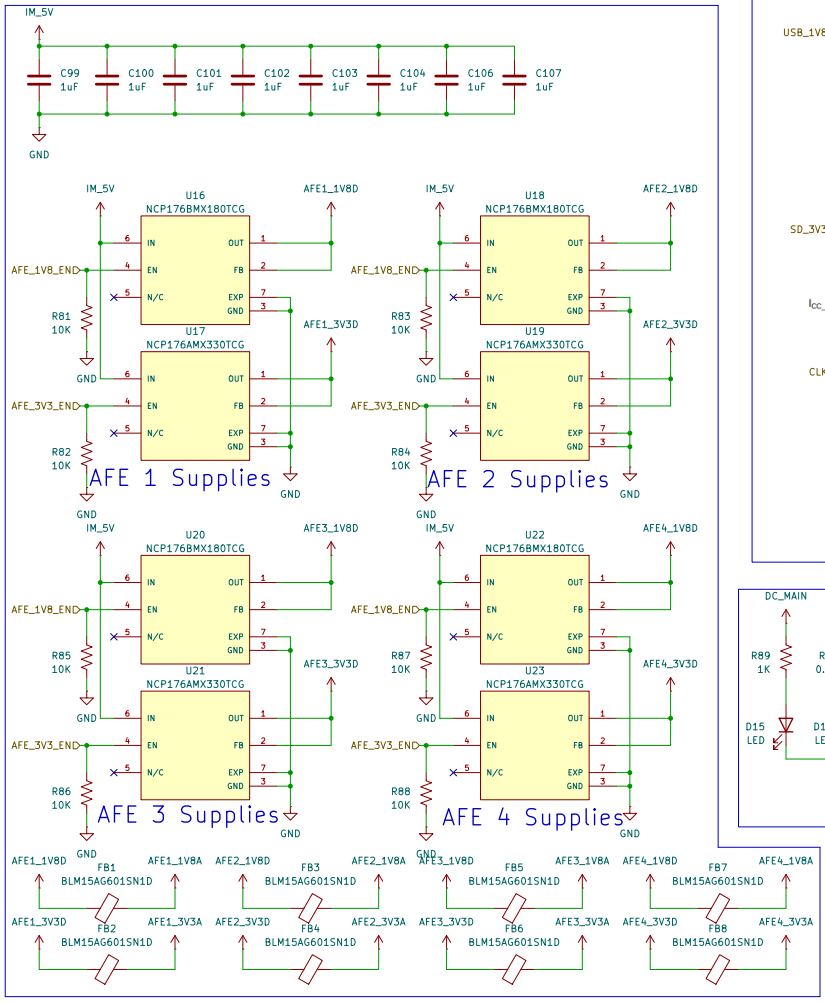
Pulser −5V Supply



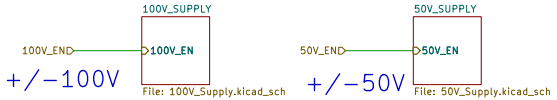


Processing Section (PL) Supplies

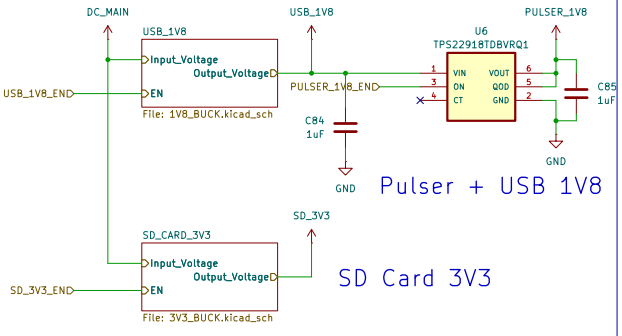


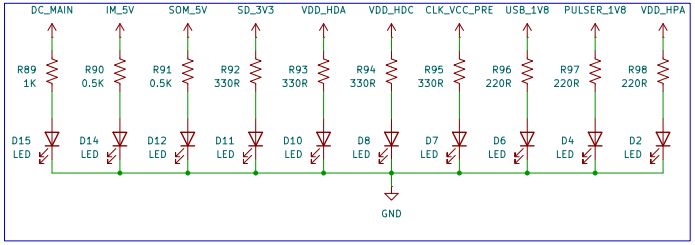


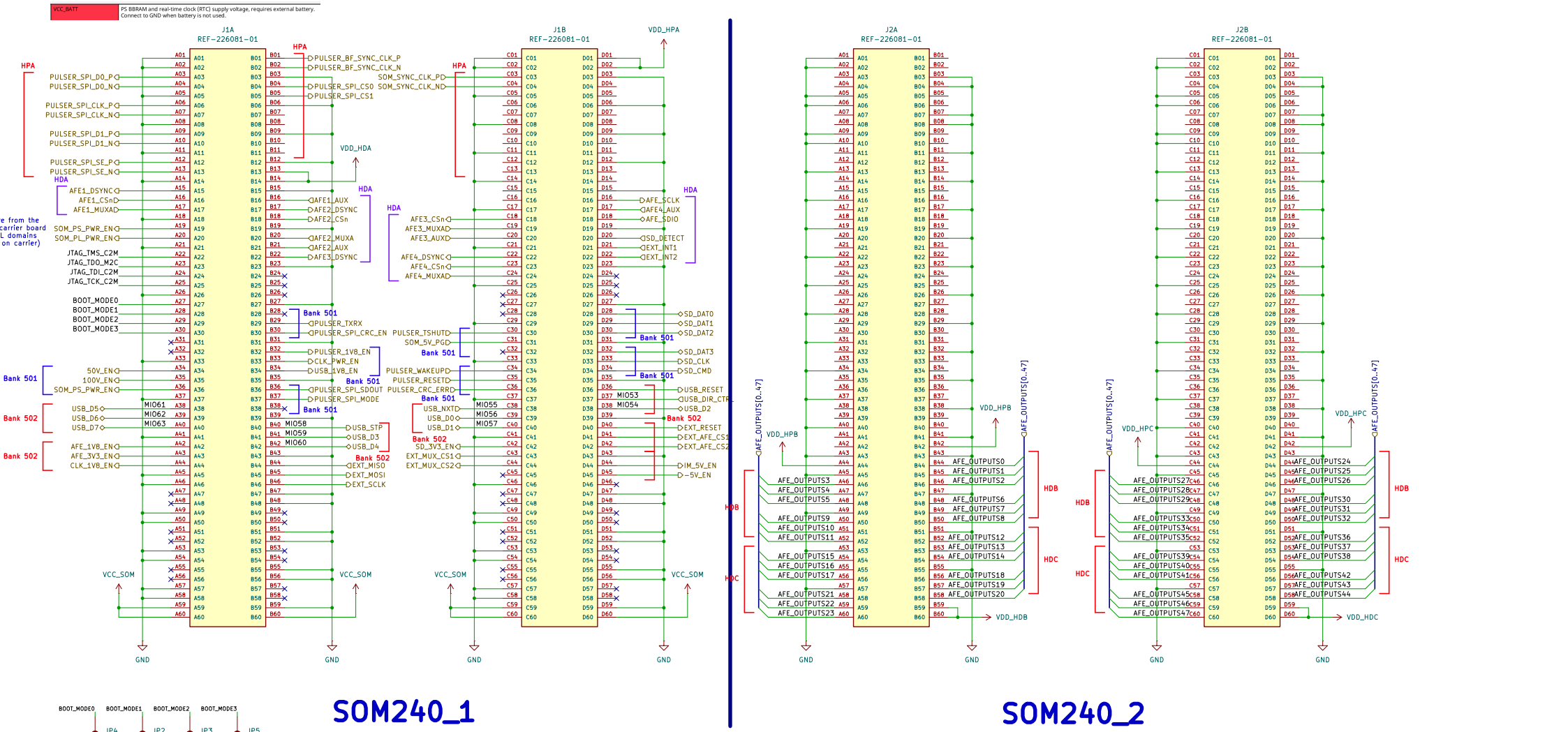
HV Supplies (+/− 50/100V)

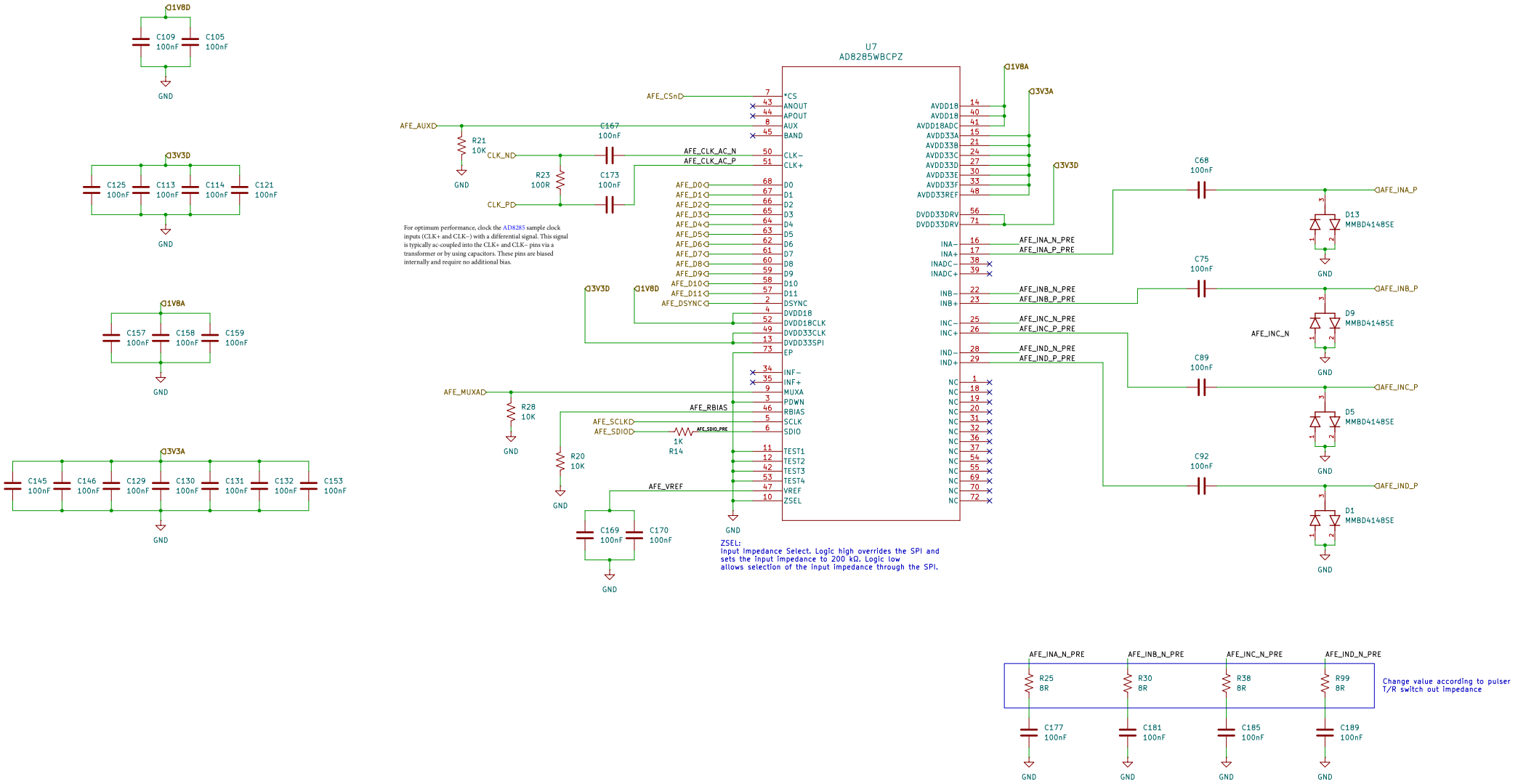


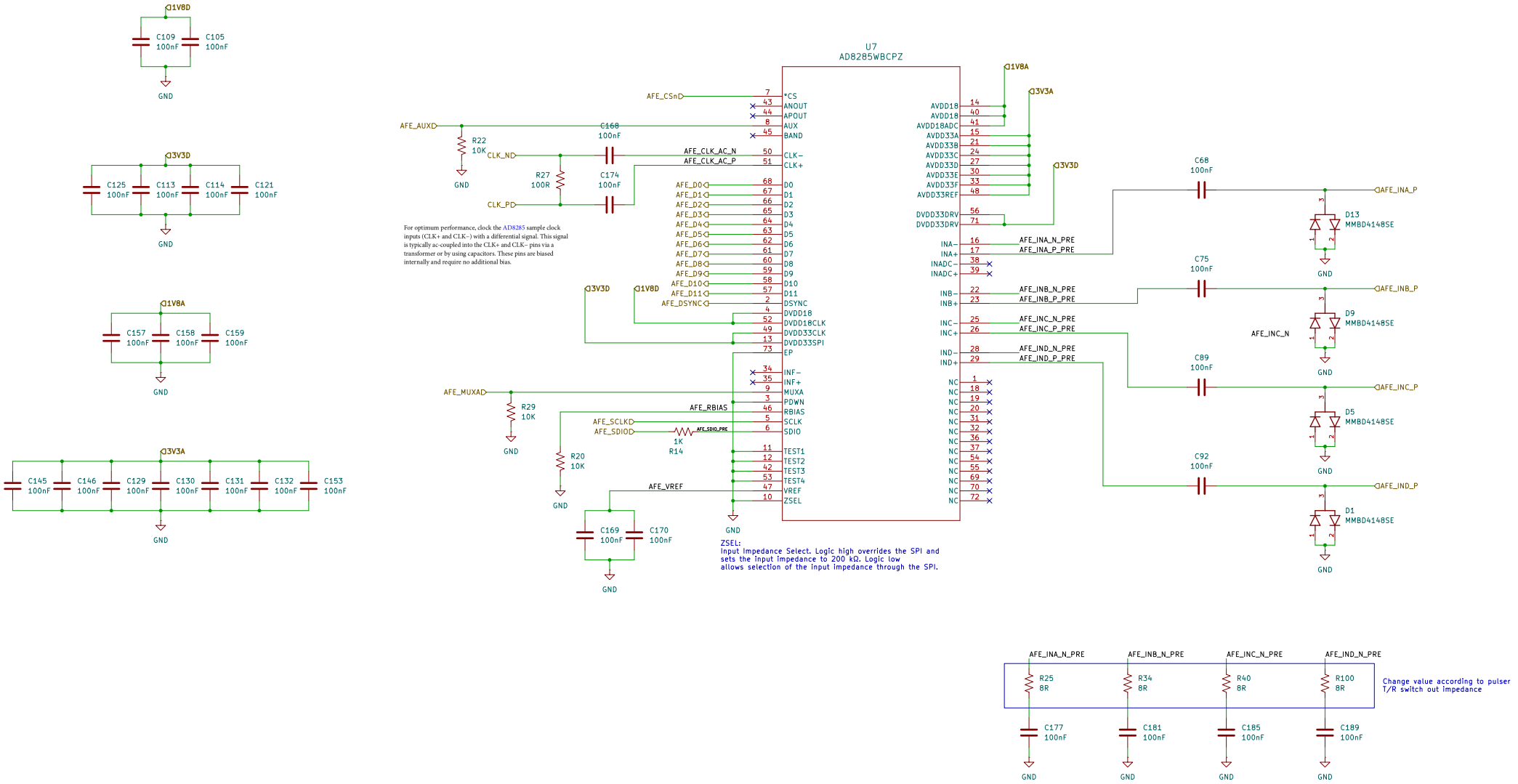
Miscellaneous Supplies

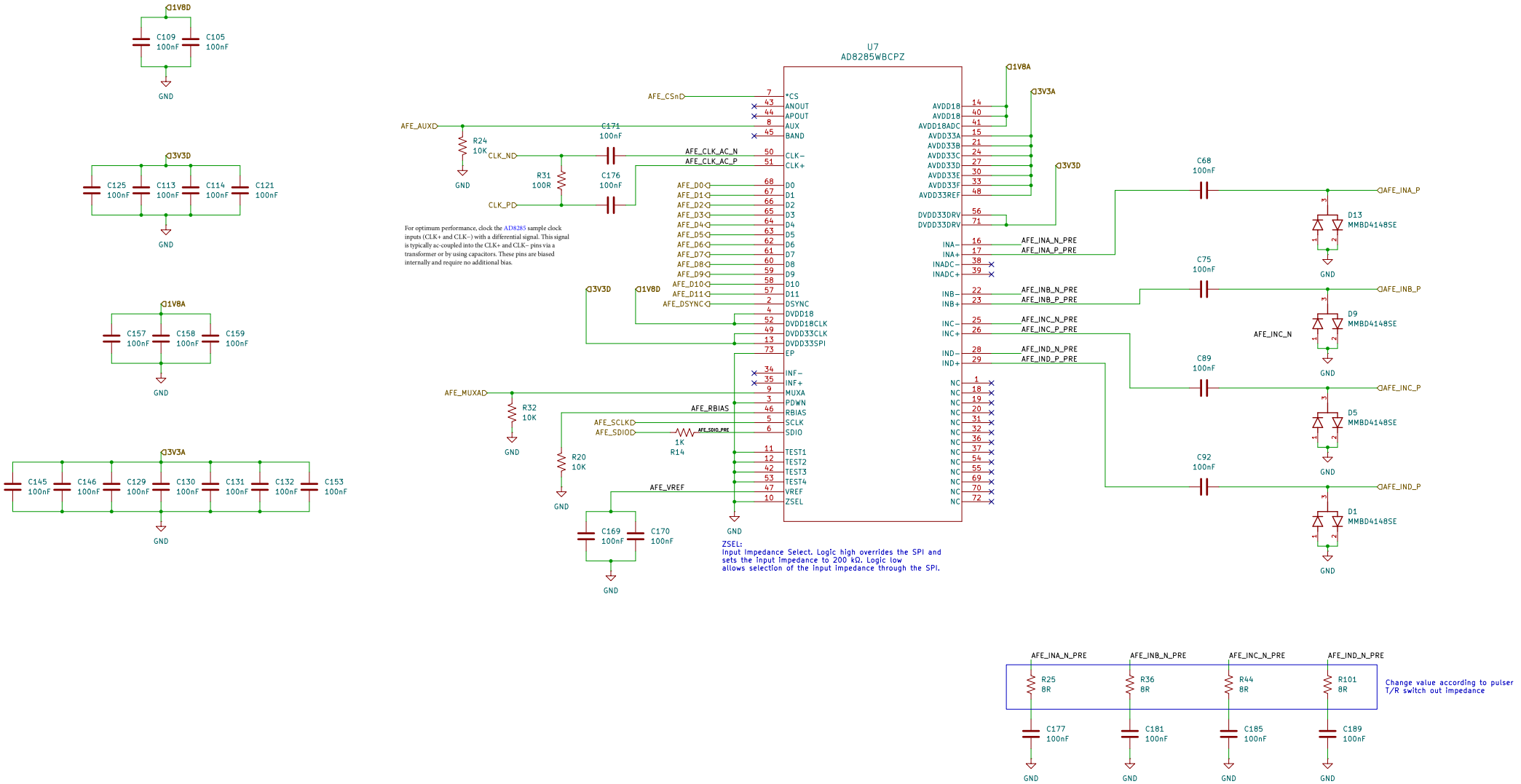


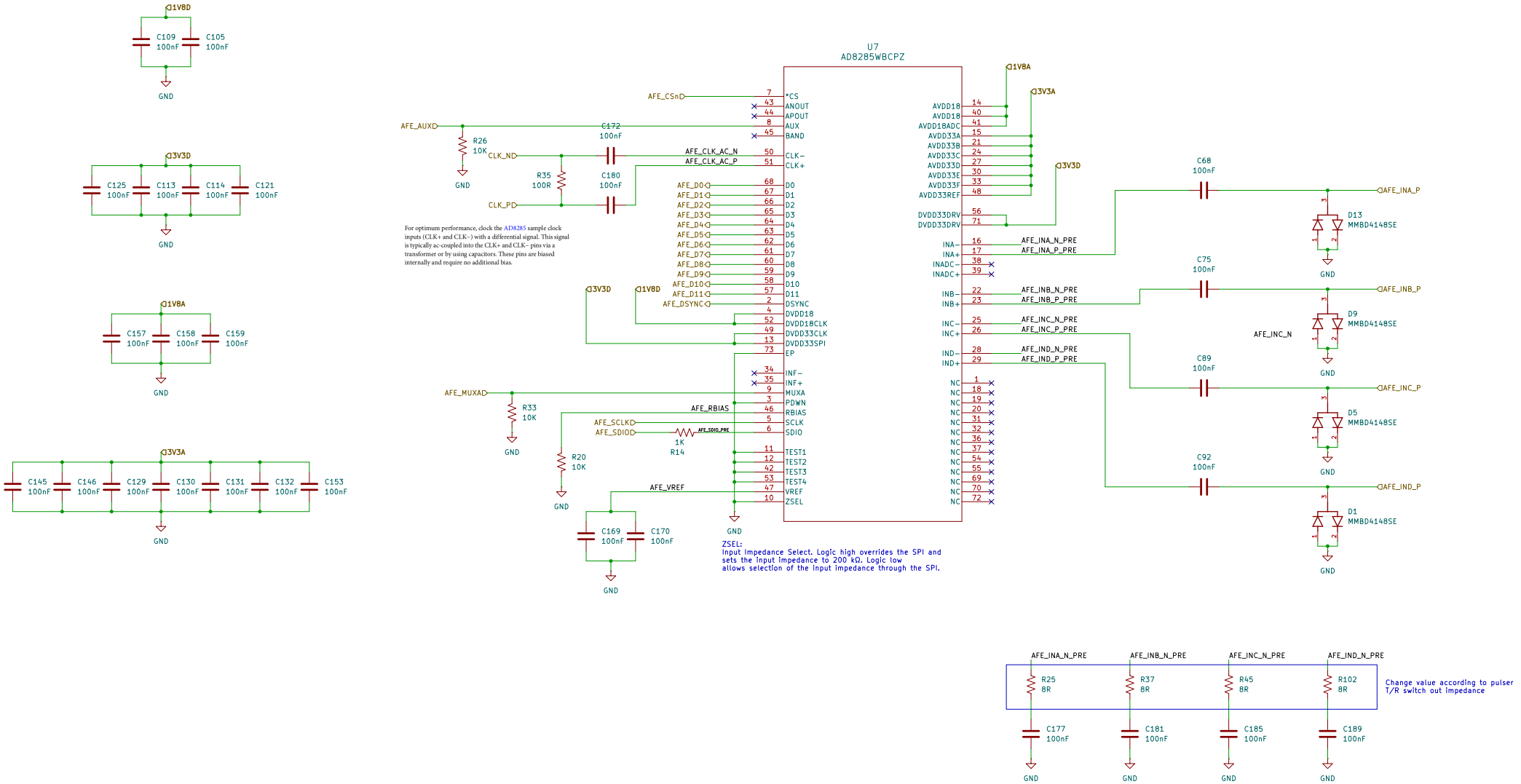



VCC_BATT
PS BBRAM and real-time clock (RTC) supply voltage, requires external battery.
Connect to GND when battery is not used.
J1A
REF-226081-01
J1B
REF-226081-01
J2A
REF-226081-01
J2B
REF-226081-01
SOM240_1
SOM240_2
BOOT_MODE0
BOOT_MODE1
BOOT_MODE2
BOOT_MODE3
JP4 BM0
JP2 BM1
JP3 BM2
JP5 BM3
Boot Mode
PS_Mode Pins[3:0]
Physical Pin Location
Quad-SPI (32 bit)
eMMC
VCC_SOM
VDD_HPA
VDD_HDA
VDD_HPB
VDD_HDB
VDD_HPC
VDD_HDC
GND
IM_5V_PGD


1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_CSn
AFE_AUX
R21 10K
C167 100nF
CLK_ND
R23 100R
C173 100nF
CLK_PD
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
AFE_MUXA
R28 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
R25 8R
R30 8R
R38 8R
R99 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF


1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R22 10K
GND
CLK_ND
CLK_PD
R27 100R
C168 100nF
C174 100nF
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_MUXA
R29 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R34 8R
R40 8R
R100 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF

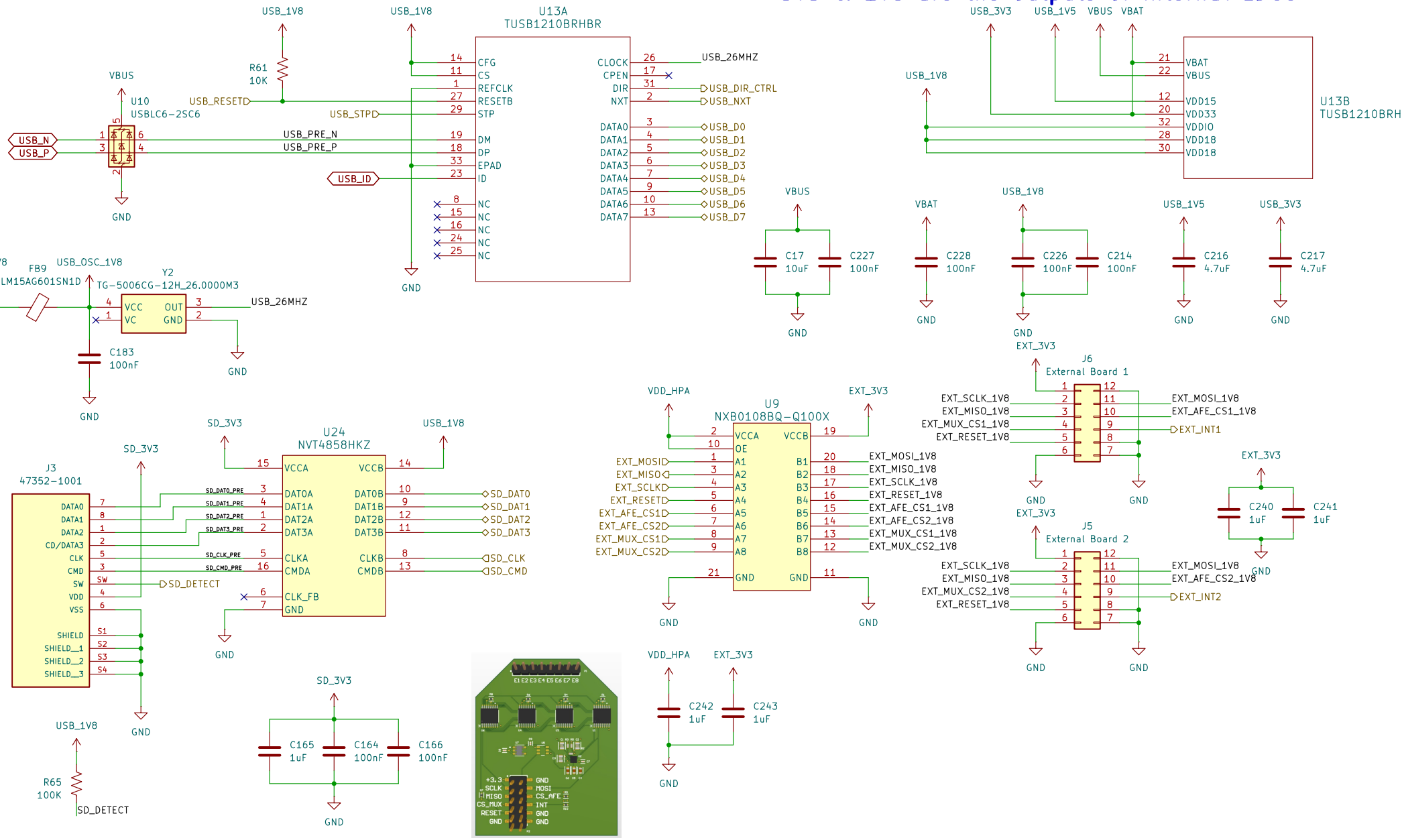

1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R24 10K
GND
CLK_ND
CLK_PD
R31 100R
C171 100nF
C176 100nF
AFE_CLK_AC_N
AFE_CLK_AC_P
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
3V3D
1V8D
AFE_MUXA
R32 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
1V8A
3V3A
3V3D
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R36 8R
R44 8R
R101 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF
GND

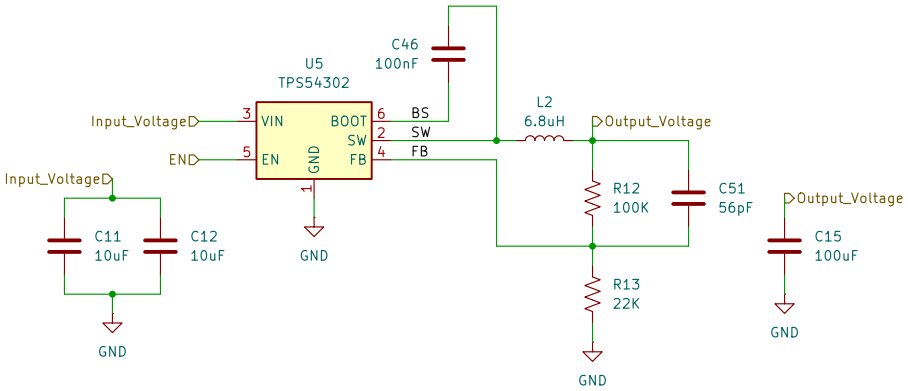

1V8D
C109 100nF
C105 100nF
GND
3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
1V8A
C157 100nF
C158 100nF
C159 100nF
GND
3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
R26 10K
GND
CLK_ND
CLK_PD
R35 100R
C172 100nF
C180 100nF
AFE_CLK_AC_N
AFE_CLK_AC_P
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
For optimum performance, clock the AD8285 sample clock inputs (CLK+ and CLK−) with a differential signal. This signal is typically ac-coupled into the CLK+ and CLK− pins via a transformer or by using capacitors. These pins are biased internally and require no additional bias.
AFE_MUXA
R33 10K
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
1K
R14
R20 10K
AFE_VREF
C169 100nF
C170 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C68 100nF
C75 100nF
C89 100nF
C92 100nF
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
AFE_INA_P
AFE_INB_P
AFE_INC_P
AFE_IND_P
AFE_INC_N
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 8R
R37 8R
R45 8R
R102 8R
Change value according to pulser T/R switch out impedance
C177 100nF
C181 100nF
C185 100nF
C189 100nF
GND

CFG	14	D	I	V_{DDIO}	REFCLK clock frequency configuration pin. Two frequencies are supported: 19.2 MHz when 0, or 26 MHz when 1.
CS	11	D	I	V_{DDIO}	Active-high chip select pin. When low the IC is in power down and ULPI bus is tri-stated. When high normal operation. Tie to V_{DDIO} if unused.

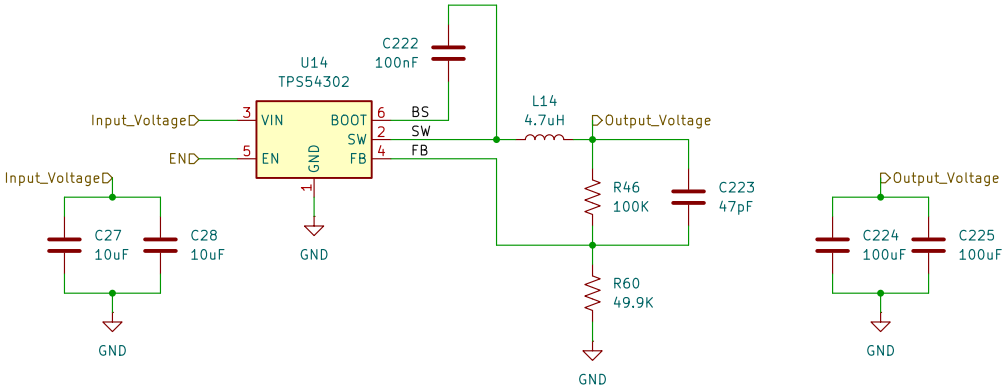
Check AN1310 for layout guidelines

3V3 & 1V5 are the outputs of internal LDOs





WEBENCH DESIGNER LINK (3.3V)



WEBENCH DESIGNER LINK (1.8V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

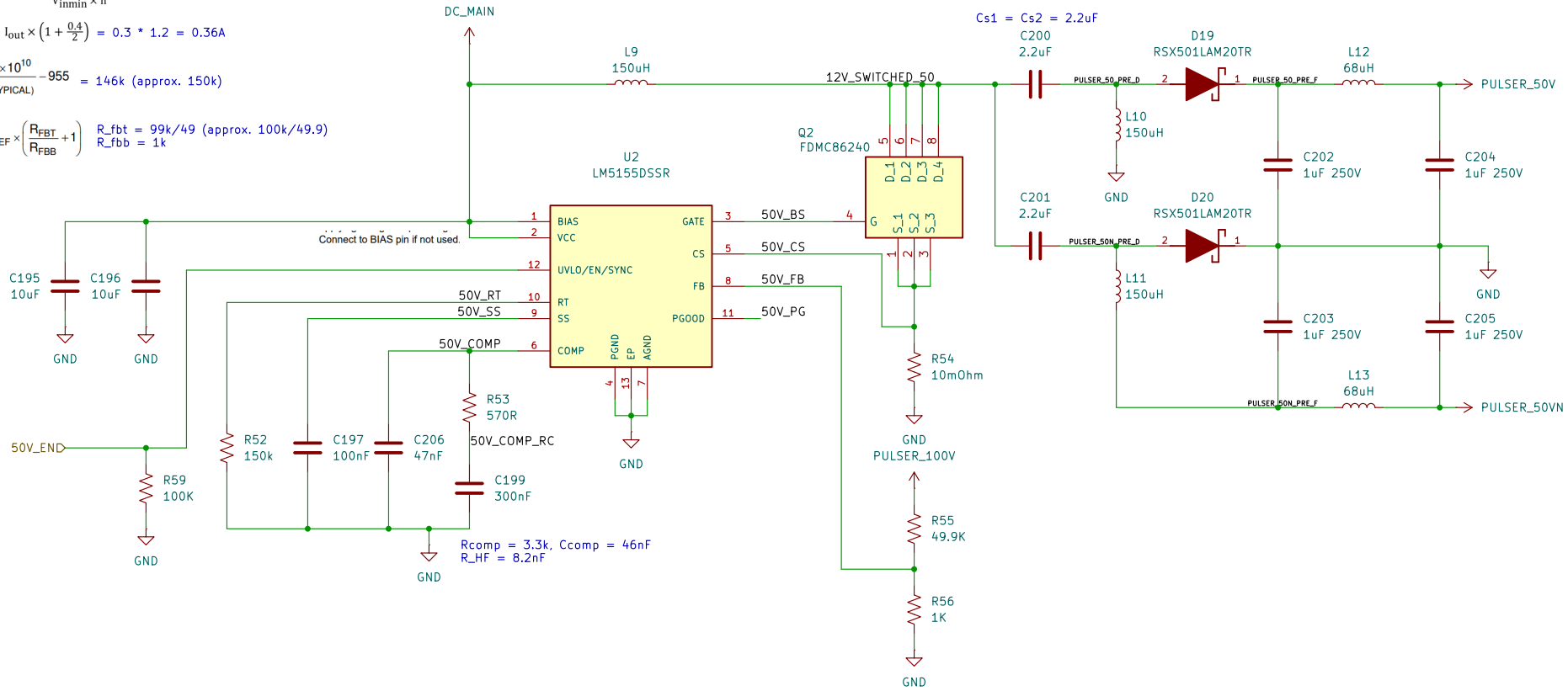
n = 0.88

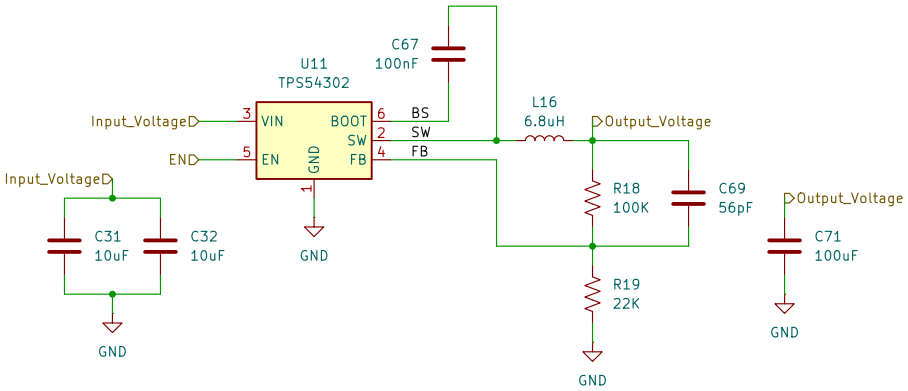
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k





WEBENCH DESIGNER LINK (3.3V)

Duty Cycle

$$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$$ = (100+0.7) / (11+100+0.7) = 90%

= (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

n = 0.88

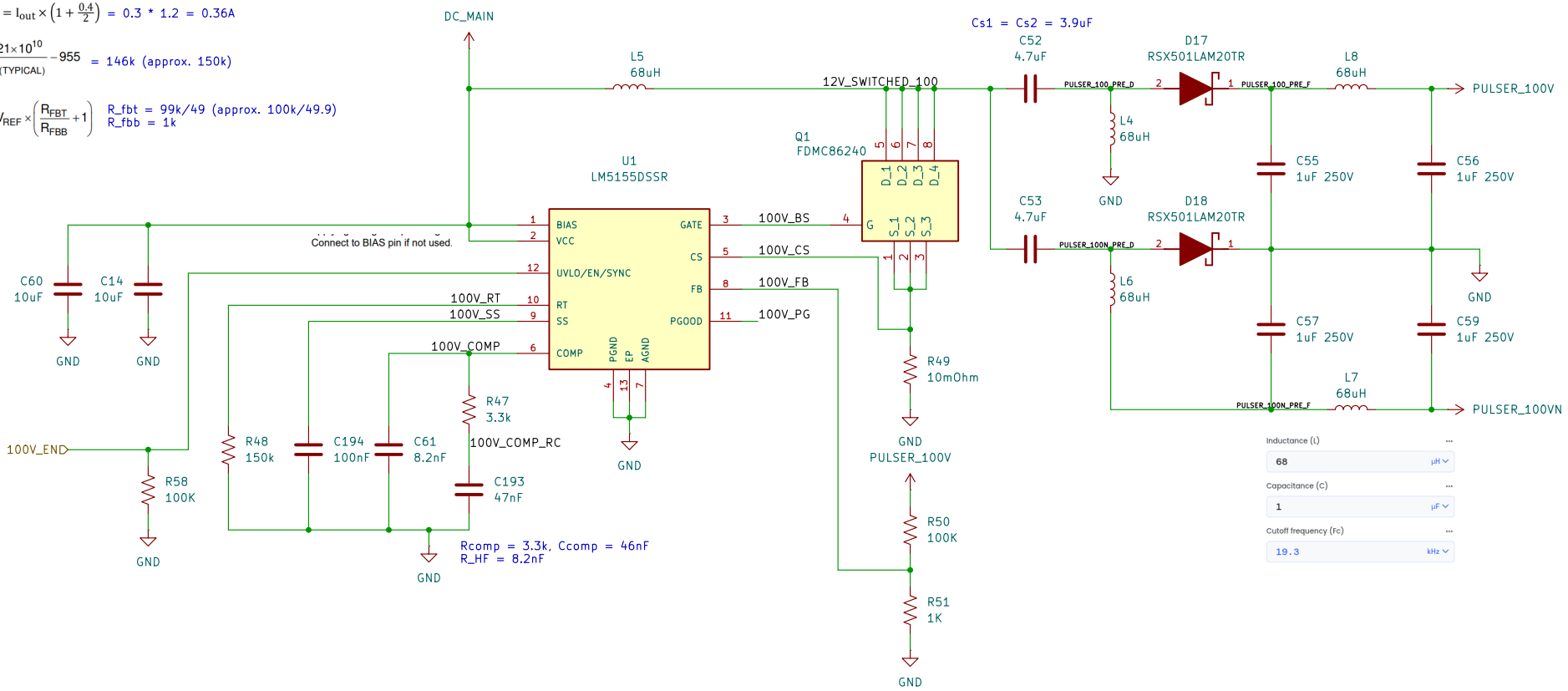
$$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

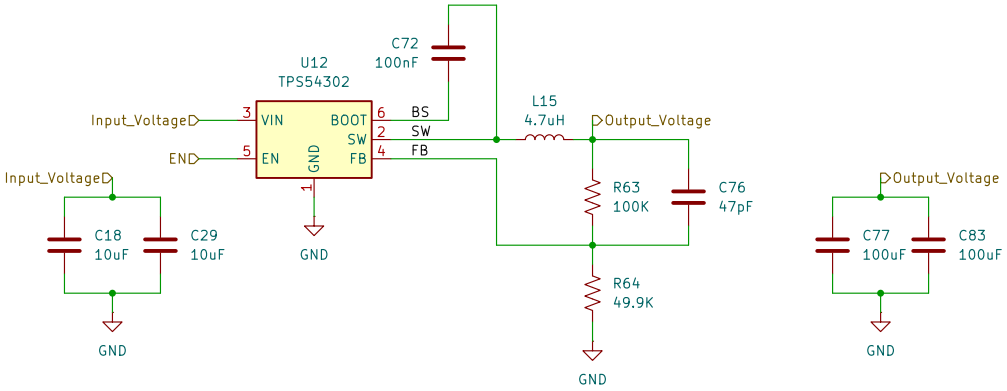
$$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$$ = 0.3 * 1.2 = 0.36A

$$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$$ = 146k (approx. 150k)

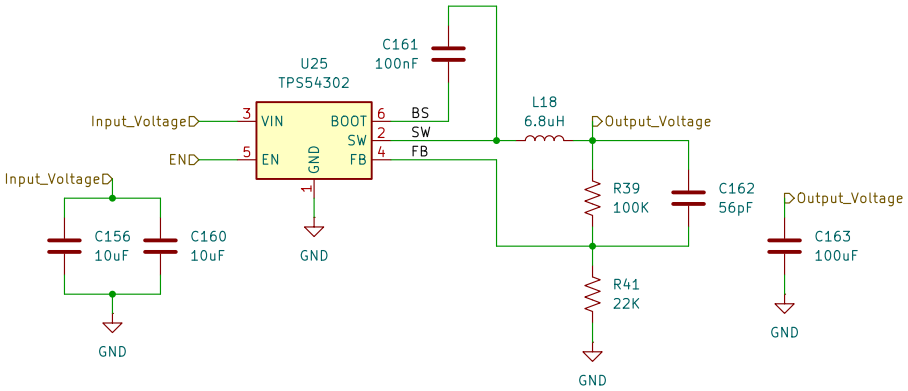
$$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$$ R_fbt = 99k/49 (approx. 100k/49.9)

R_fbb = 1k

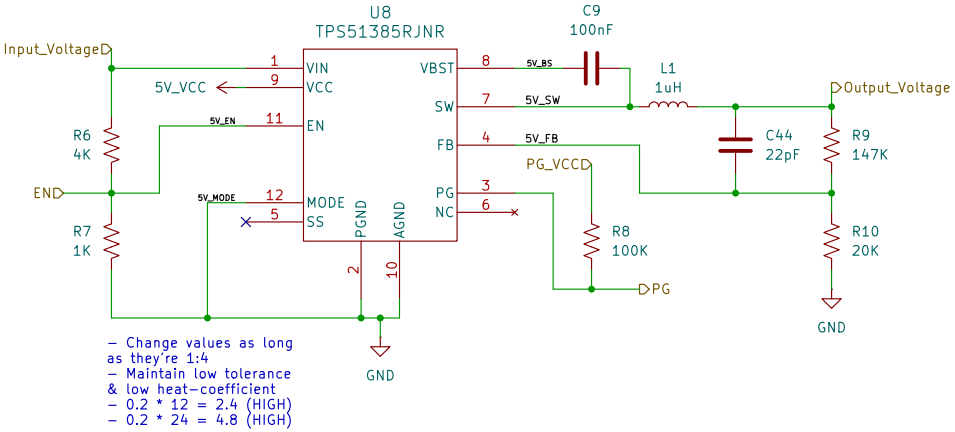


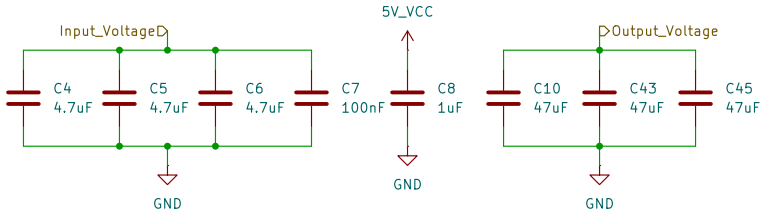


WEBENCH DESIGNER LINK (1.8V)

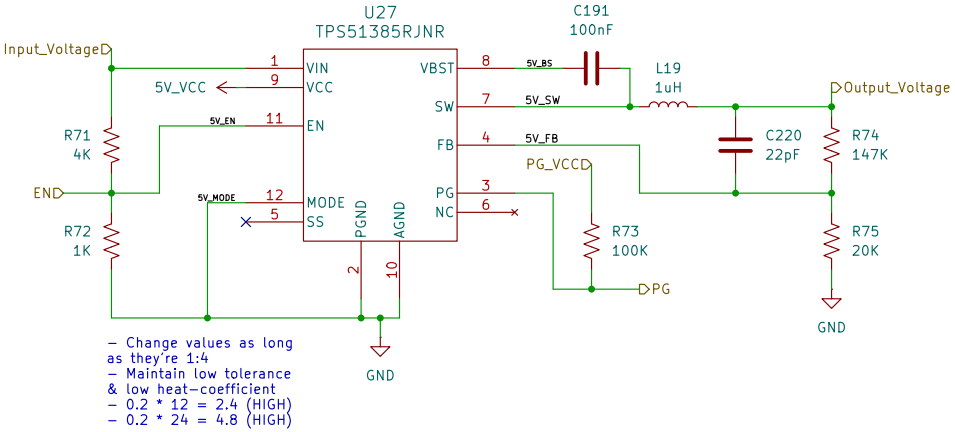


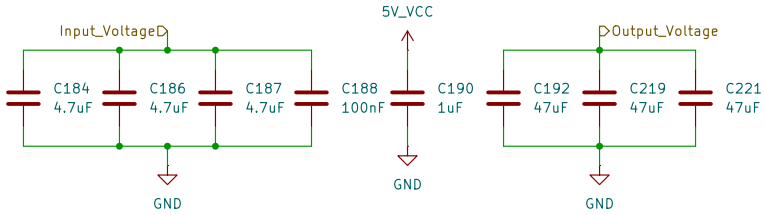
WEBENCH DESIGNER LINK (3.3V)





WeBench Designer Link





WeBench Designer Link